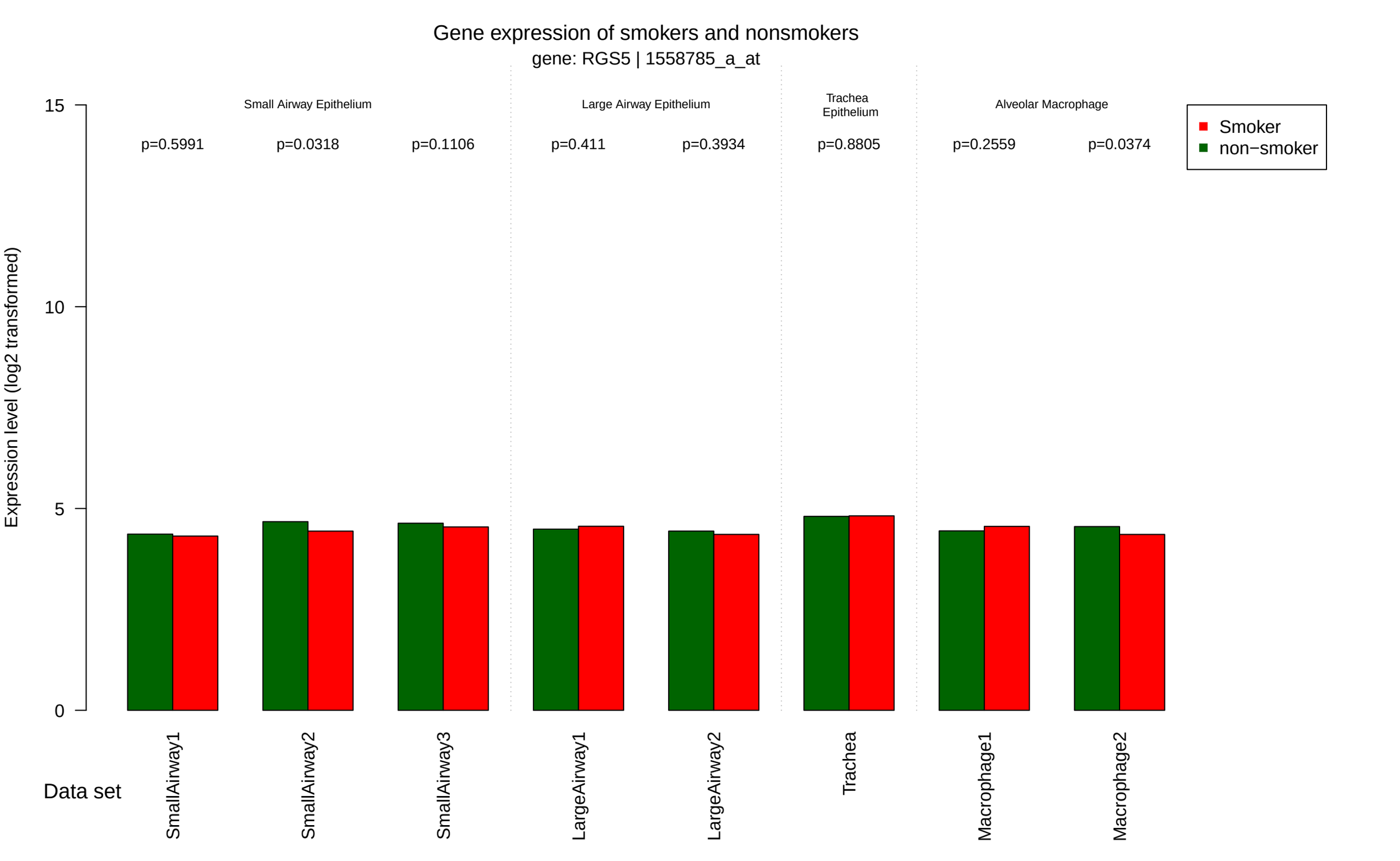
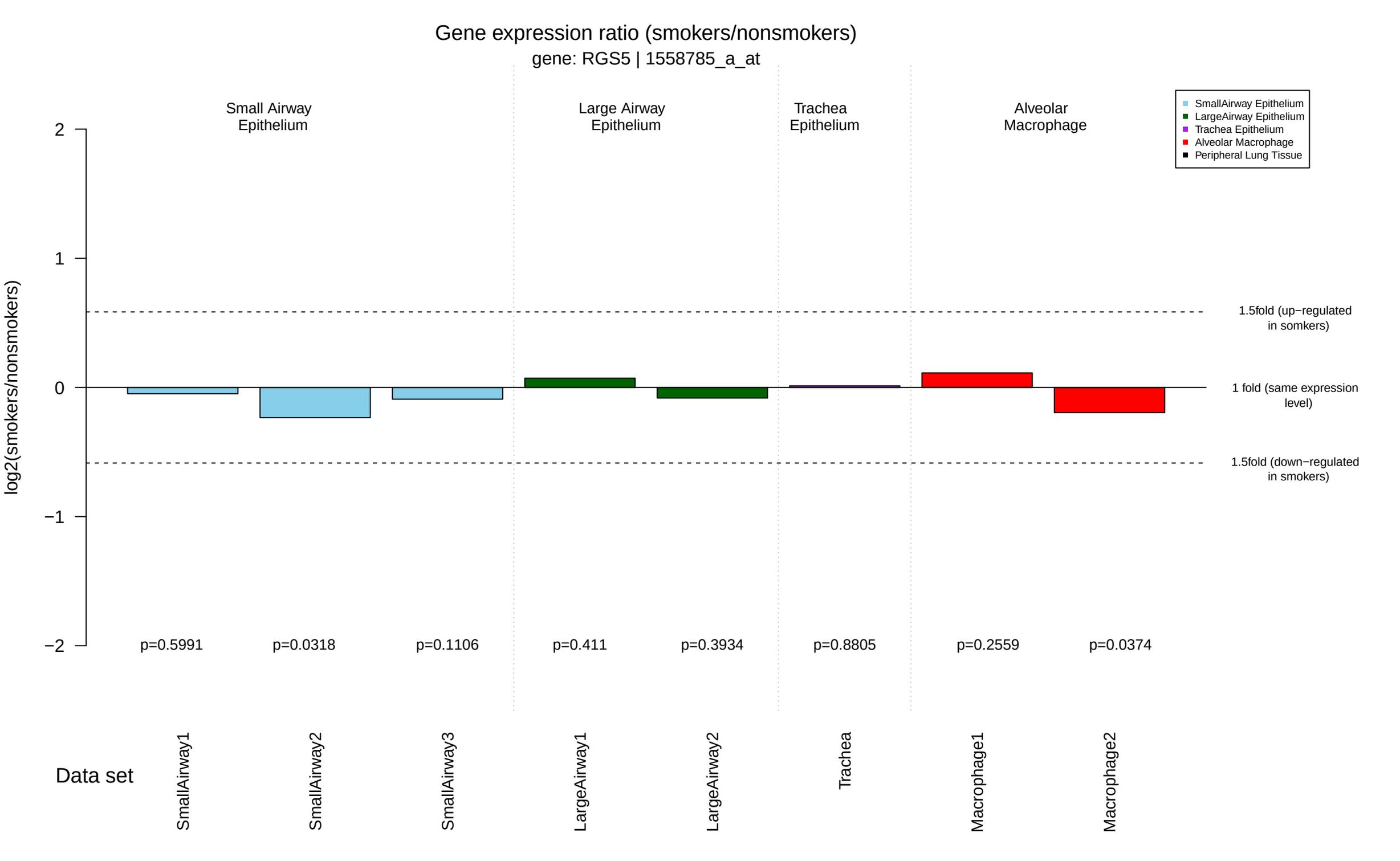
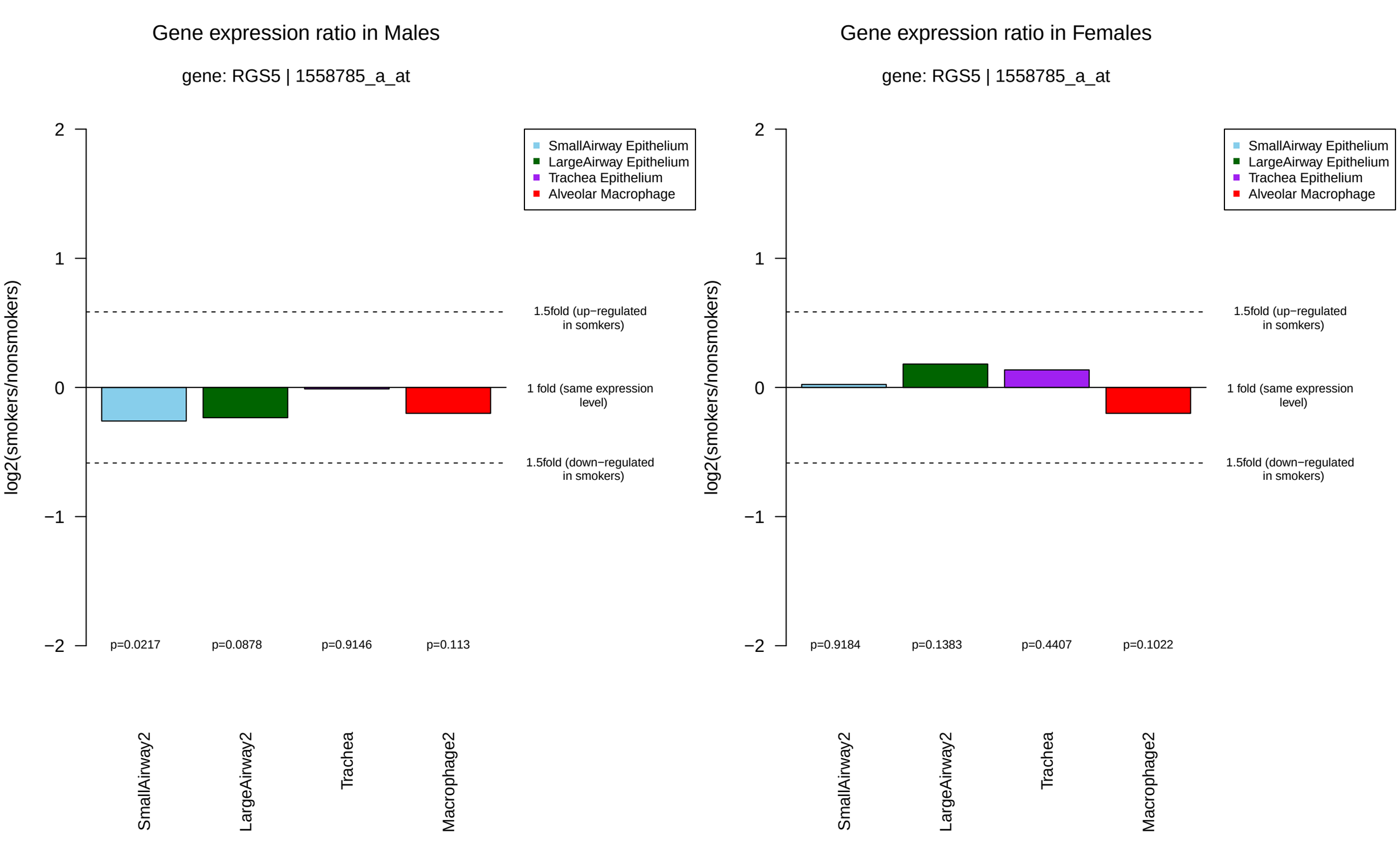
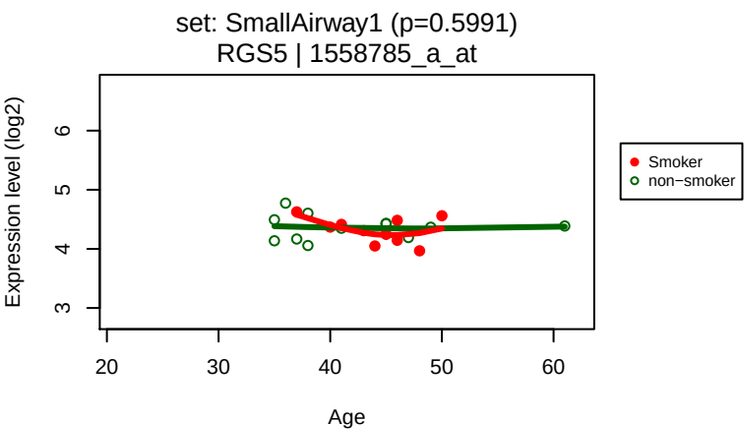
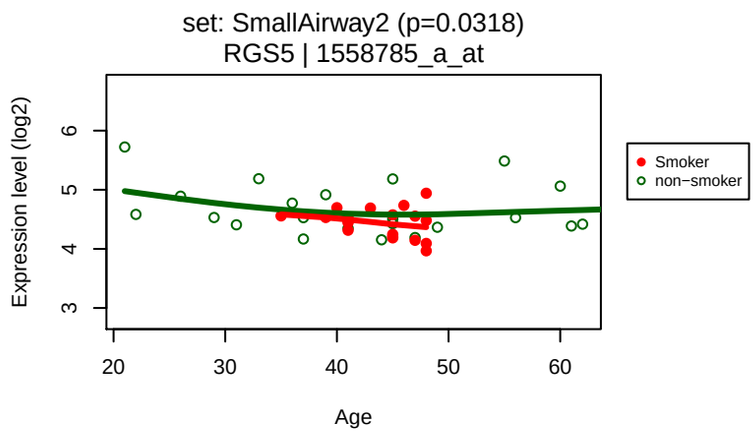
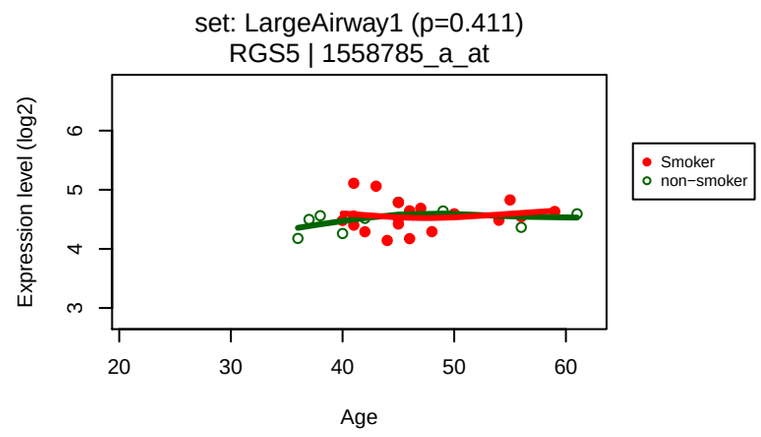

Gene expression of smokers and nonsmokers
gene: RGS5 | 1558785_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5991
p=0.0318
p=0.1106
p=0.411
p=0.3934
p=0.8805
p=0.2559
p=0.0374
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

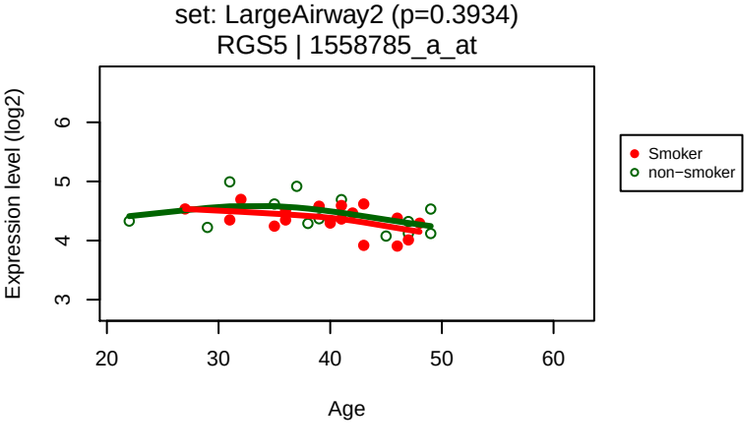

Gene expression ratio (smokers/nonsmokers)
gene: RGS5 | 1558785_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5991
p=0.0318
p=0.1106
p=0.411
p=0.3934
p=0.8805
p=0.2559
p=0.0374
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

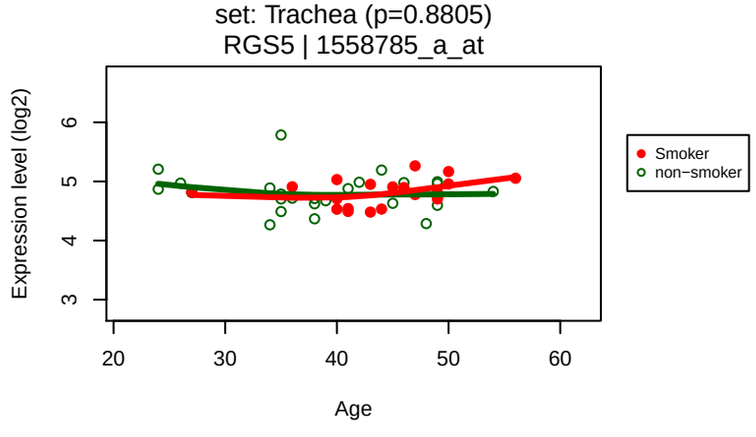

Gene expression ratio in Males
gene: RGS5 | 1558785_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0217
p=0.0878
p=0.9146
p=0.113
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS5 | 1558785_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9184
p=0.1383
p=0.4407
p=0.1022
SmallAirway2
LargeAirway2
Trachea
Macrophage2

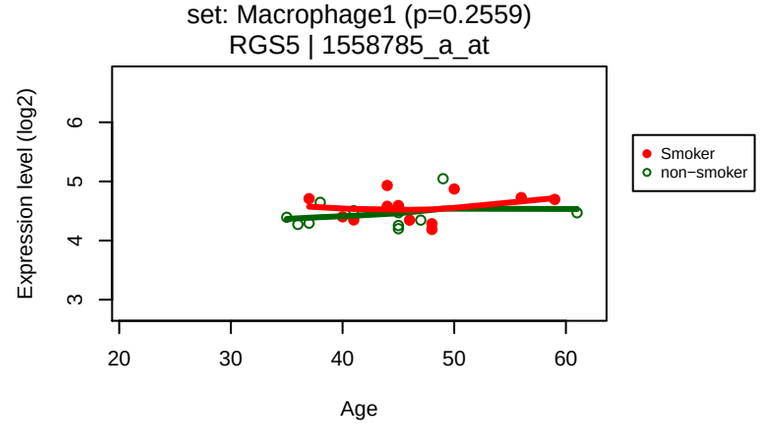

set: SmallAirway1 (p=0.5991)
RGS5 | 1558785_a_at
Expression level (log2)
Age
Smoker
non−smoker

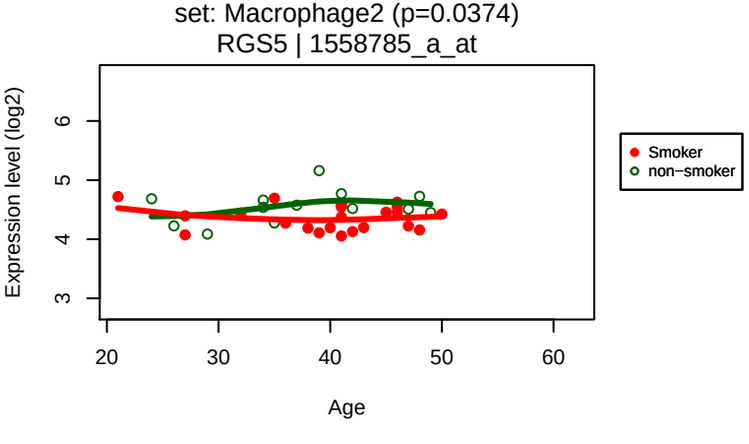

set: SmallAirway2 (p=0.0318)
RGS5 | 1558785_a_at
Expression level (log2)
Age
Smoker
non−smoker

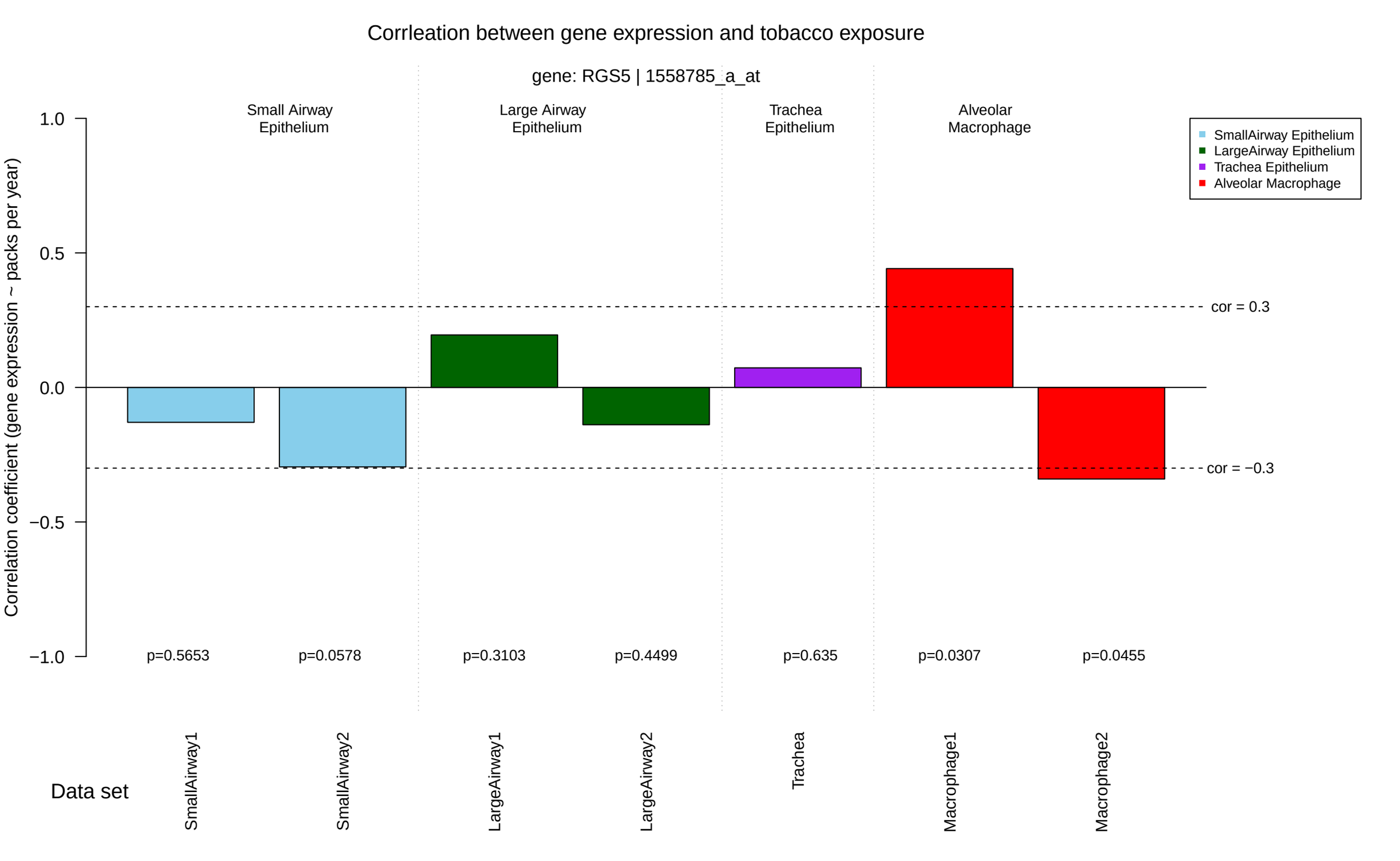

set: LargeAirway1 (p=0.411)
RGS5 | 1558785_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3934)
RGS5 | 1558785_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8805)
RGS5 | 1558785_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2559)
RGS5 | 1558785_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0374)
RGS5 | 1558785_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS5 | 1558785_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5653
p=0.0578
p=0.3103
p=0.4499
p=0.635
p=0.0307
p=0.0455
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2